Harmonic Motion

- A mass m on a spring of stiffness k oscillates.
- The angular frequency is $\omega = \sqrt{k/m}$.
- The period is $T = 2\pi/\omega$.

The Equation of Motion

Newton's second law for the spring gives

$$m\ddot{x} = -kx.$$

Its solution is

$$x(t) = A\cos(\omega t + \phi).$$

Energy

The two forms of energy are

$$K = \tfrac{1}{2}mv^2,$$
$$U = \tfrac{1}{2}kx^2,$$

and their sum $E = K + U$ stays constant.